



TECNOLÓGICO
NACIONAL DE MÉXICO



Tecnológico Nacional de México

Campus Saltillo

Ingeniería en Sistemas Computacionales

Computo en la nube

4.10 – Análisis de tráfico de red con Wireshark

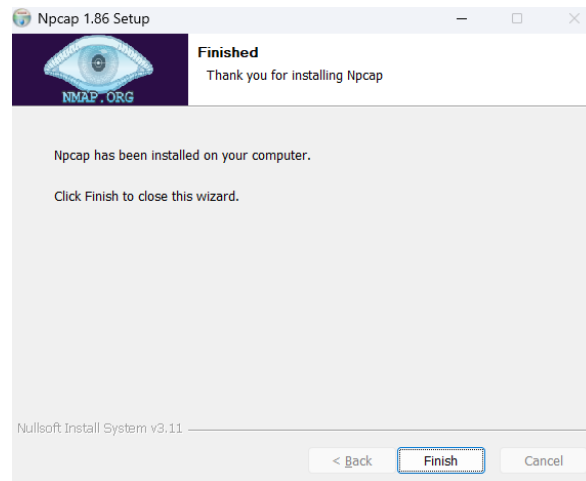
Docente: Ing. Miguel Salazar del Bosque

Alumno:

Jesus Contreras Castillo 22050648

25/03/2026}

1- Instalación de Wireshark



2- Después de analizar el tráfico de la red, filtrar por http

No.	Time	Source	Destination	Protocol	Length	Info
230	13.920699500	192.168.1.78	52.217.134.173	HTTP	352	GET /SSL.com-TLS-T-ECC-R2.cer HTTP/1.1
239	14.067348900	52.217.134.173	192.168.1.78	HTTP	878	HTTP/1.1 200 OK (application/pkix-cert)

3- Filtrar por DNS

No.	Time	Source	Destination	Protocol	Length	Info
129	13.341558000	fe80::5601:7e18:84c...	fe80::1	DNS	91	Standard query 0xd4f5 HTTPS example.com
130	13.343464700	fe80::5601:7e18:84c...	fe80::1	DNS	91	Standard query 0x758b AAAA example.com
131	13.344302800	fe80::5601:7e18:84c...	fe80::1	DNS	91	Standard query 0x8c85 A example.com
132	13.360989400	fe80::1	fe80::5601:7e18:84c...	DNS	249	Standard query response 0xd4f5 HTTPS example.com HTTPS A 104.18.26.120 A 104.18.27.120 AAAA 2606:4700::6812:1b78 AAAA 2606:4700::6812:1a...
133	13.361559500	fe80::1	fe80::5601:7e18:84c...	DNS	147	Standard query response 0x758b AAAA example.com AAAA 2606:4700::6812:1a78 AAAA 2606:4700::6812:1b78
134	13.362809000	fe80::1	fe80::5601:7e18:84c...	DNS	123	Standard query response 0x8c85 A example.com A 104.18.27.120 A 104.18.26.120
135	13.608070200	fe80::5601:7e18:84c...	fe80::1	DNS	115	Standard query 0x873f HTTPS optimizationguide-pa.googleapis.com
136	13.607343100	fe80::5601:7e18:84c...	fe80::1	DNS	115	Standard query 0xeea4 AAAA optimizationguide-pa.googleapis.com
137	13.607584700	fe80::5601:7e18:84c...	fe80::1	DNS	115	Standard query 0x093d A optimizationguide-pa.googleapis.com
139	13.626809400	fe80::5601:7e18:84c...	fe80::1	DNS	91	Standard query 0x170e HTTPS example.com
140	13.627351600	fe80::5601:7e18:84c...	fe80::1	DNS	91	Standard query 0xf030 AAAA example.com
141	13.627660800	fe80::5601:7e18:84c...	fe80::1	DNS	91	Standard query 0xe4e2 A example.com
144	13.630754400	fe80::1	fe80::5601:7e18:84c...	DNS	172	Standard query response 0x873f HTTPS optimizationguide-pa.googleapis.com SOA ns1.google.com
145	13.630754400	fe80::1	fe80::5601:7e18:84c...	DNS	179	Standard query response 0x093d A optimizationguide-pa.googleapis.com A 216.239.38.223 A 216.239.36.223 A 216.239.34.223 A 216.239.32.223
146	13.630754400	fe80::1	fe80::5601:7e18:84c...	DNS	227	Standard query response 0xeea4 AAAA optimizationguide-pa.googleapis.com AAAA 2001:4860:4802:38::1223 AAAA 2001:4860:4802:36::1223 AAAA 2001:4860:4802:35::1223
147	13.630754400	fe80::1	fe80::5601:7e18:84c...	DNS	147	Standard query response 0xf030 AAAA example.com AAAA 2606:4700::6812:1b78 AAAA 2606:4700::6812:1a78
148	13.630754400	fe80::1	fe80::5601:7e18:84c...	DNS	123	Standard query response 0xe4e2 A example.com A 104.18.26.120 A 104.18.27.120
154	13.646182800	fe80::1	fe80::5601:7e18:84c...	DNS	249	Standard query response 0x170e HTTPS example.com HTTPS A 104.18.26.120 A 104.18.27.120 AAAA 2606:4700::6812:1b78 AAAA 2606:4700::6812:1a...
184	13.723800000	fe80::5601:7e18:84c...	fe80::1	DNS	92	Standard query 0x9e2b HTTPS cert.ssl.com
185	13.723544100	fe80::5601:7e18:84c...	fe80::1	DNS	92	Standard query 0xb138 AAAA cert.ssl.com
186	13.723580800	fe80::5601:7e18:84c...	fe80::1	DNS	92	Standard query 0x3dfc A cert.ssl.com
191	13.746331900	fe80::1	fe80::5601:7e18:84c...	DNS	174	Standard query response 0x9e2b HTTPS cert.ssl.com SOA ns-1026.ausdns-00.org
192	13.746331900	fe80::1	fe80::5601:7e18:84c...	DNS	174	Standard query response 0xb138 AAAA cert.ssl.com SOA ns-1026.ausdns-00.org

4- Filtrar tcp

No.	Time	Source	Destination	Protocol	Length	Info
206	13.781449000	2806:108e:30:8af9:a...	2a06:98c1:310b::ac4...	TLSv1.2	957	Application Data
208	13.802240000	192.168.1.78	52.217.134.173	TCP	66	50060 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
209	13.808171200	2a06:98c1:310b::ac4...	2806:108e:30:8af9:a...	TLSv1.2	153	Application Data
213	13.815046900	2a06:98c1:310b::ac4...	2806:108e:30:8af9:a...	TCP	74	443 → 59054 [ACK] Seq=1864 Ack=1279 Win=19 Len=0
217	13.851133900	2806:108e:30:8af9:a...	2a06:98c1:310b::ac4...	TCP	74	61640 → 443 [ACK] Seq=181 Ack=159 Win=251 Len=0
218	13.854105100	2a06:98c1:310b::ac4...	2806:108e:30:8af9:a...	TLSv1.2	415	Application Data
219	13.854105100	2a06:98c1:310b::ac4...	2806:108e:30:8af9:a...	TLSv1.2	121	Application Data
220	13.854105100	2a06:98c1:310b::ac4...	2806:108e:30:8af9:a...	TLSv1.2	105	Application Data
221	13.854148300	2806:108e:30:8af9:a...	2a06:98c1:310b::ac4...	TCP	74	63436 → 443 [ACK] Seq=33171 Ack=1671 Win=252 Len=0
228	13.919711000	52.217.134.173	192.168.1.78	TCP	66	80 → 50060 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1432 WS=256 SACK_PERM
229	13.919932300	192.168.1.78	52.217.134.173	TCP	54	50060 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
230	13.920699500	192.168.1.78	52.217.134.173	HTTP	352	GET /SSL.com-TLS-T-ECC-R2.cer HTTP/1.1
231	13.926428400	2a06:98c1:310b::ac4...	2806:108e:30:8af9:a...	TLSv1.2	650	Application Data
232	13.926428400	2a06:98c1:310b::ac4...	2806:108e:30:8af9:a...	TLSv1.2	121	Application Data
233	13.926428400	2a06:98c1:310b::ac4...	2806:108e:30:8af9:a...	TLSv1.2	105	Application Data
234	13.926543200	2806:108e:30:8af9:a...	2a06:98c1:310b::ac4...	TCP	74	59054 → 443 [ACK] Seq=1279 Ack=1718 Win=253 Len=0
235	13.928276200	2806:108e:30:8af9:a...	2a06:98c1:310b::ac4...	TCP	109	Application Data
236	14.004035000	2a06:98c1:310b::ac4...	2806:108e:30:8af9:a...	TCP	74	443 → 59054 [ACK] Seq=1718 Ack=1314 Win=19 Len=0
237	14.037782200	52.217.134.173	192.168.1.78	TCP	54	80 → 50060 [ACK] Seq=1 Ack=299 Win=64256 Len=0
238	14.067348900	52.217.134.173	192.168.1.78	TCP	493	80 → 50060 [PSH, ACK] Seq=1 Ack=299 Win=64256 Len=439 [TCP PDU reassembled in 239]
239	14.067348900	52.217.134.173	192.168.1.78	HTTP	878	HTTP/1.1 200 OK (application/pkix-cert)
240	14.067474500	192.168.1.78	52.217.134.173	TCP	54	50060 → 80 [ACK] Seq=299 Ack=1264 Win=64256 Len=0
241	14.082779500	2806:108e:30:8af9:a...	2a06:98c1:310b::ac4...	TLSv1.2	138	Change Cipher Spec, Application Data

Frame 230: Packet, 352 bytes on wire (464 bits), 352 bytes captured (464 bits) on interface \Device\NPF... (36f51...)

Ethernet II, Src: Intel18:2a:4f (70:1a:b8:18:2a:4f)

Internet Protocol Version 4, Src: 213.227.173.141, Dst: 192.168.1.78

Transmission Control Protocol, Src Port: 443, Dst Port: 56768, Seq: 25, Ack: 49, Len: 0

5- ¿Qué puedo ver en un http?

No.	Time	Source	Destination	Protocol	Length	Info
208	13.802240000	192.168.1.78	52.217.134.173	TCP	66	50060 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
228	13.919711000	52.217.134.173	192.168.1.78	TCP	66	80 → 50060 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1432 WS=256 SACK_PERM
229	13.919932300	192.168.1.78	52.217.134.173	TCP	54	50060 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
230	13.920699500	192.168.1.78	52.217.134.173	HTTP	352	GET /SSL.com-TLS-T-ECC-R2.cer HTTP/1.1
237	14.037782200	52.217.134.173	192.168.1.78	TCP	54	80 → 50060 [ACK] Seq=1 Ack=299 Win=64256 Len=0
238	14.067348900	52.217.134.173	192.168.1.78	TCP	493	80 → 50060 [PSH, ACK] Seq=1 Ack=299 Win=64256 Len=439 [TCP PDU reassembled in 239]
239	14.067348900	52.217.134.173	192.168.1.78	HTTP	878	HTTP/1.1 200 OK (application/pkix-cert)
240	14.067474500	192.168.1.78	52.217.134.173	TCP	54	50060 → 80 [ACK] Seq=299 Ack=1264 Win=64256 Len=0
241	14.082779500	2806:108e:30:8af9:a...	2a06:98c1:310b::ac4...	TLSv1.2	138	Change Cipher Spec, Application Data

Frame 230: Packet, 352 bytes on wire (464 bits), 352 bytes captured (464 bits) on interface \Device\NPF... (36f51...)

Ethernet II, Src: Intel18:2a:4f (70:1a:b8:18:2a:4f)

Internet Protocol Version 4, Src: 213.227.173.141, Dst: 192.168.1.78

Transmission Control Protocol, Src Port: 443, Dst Port: 56768, Seq: 25, Ack: 49, Len: 0

Hypertext Transfer Protocol

GET /SSL.com-TLS-T-ECC-R2.cer HTTP/1.1

Host: cert.ssl.com

Connection: keep-alive

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36

Accept-Encoding: gzip, deflate

Accept-Language: es-es-US;q=0.9,es-419;q=0.8,en;q=0.7

HTTP/1.1 200 OK

x-amz-id-2: QJSTeeSietver3s/Rc2TolghBcTi6pEnSe37R3a2olmPnHqAQcok9bm7A5psQVeiKJ3HbYXg=

x-amz-request-id: 2Q0BAEMS1VP12NF2

Date: Thu, 26 Mar 2026 05:03:37 GMT

Cache-Control: max-age=31556952, public

Last-Modified: Thu, 27 Oct 2022 12:43:21 GMT

x-amz-version-id: 8Ynh5AiscalfIBXqHkeJH9KPTVA1sLL5

ETag: "2ef59e3b9f39fb060cd602a7343e7e6d"

Content-Type: application/pkix-cert

Content-Length: 824

Server: AmazonS3

0..40....."H.....JS...0

..*H.....001.0

..SSL Corporation100...U...SSL.com TLS ECC Root CA 20220..

2210211702232

371017170222801.0

..SSL Corporation100...U...SSL.com TLS Transit ECC CA R20b0...H.....*b..d..}..S\.....#.....c..(h.....h..w..v..P..Y(-o.6.....L"...?..z..s..

..:7..

..E..sr...e..x..l..f..g..)...YB..U0...U.....0.....U..0...../...+..T[V.&g8...0L...*.....@0>0<+.....0..0http://cert.ssl.com/SSL.com-

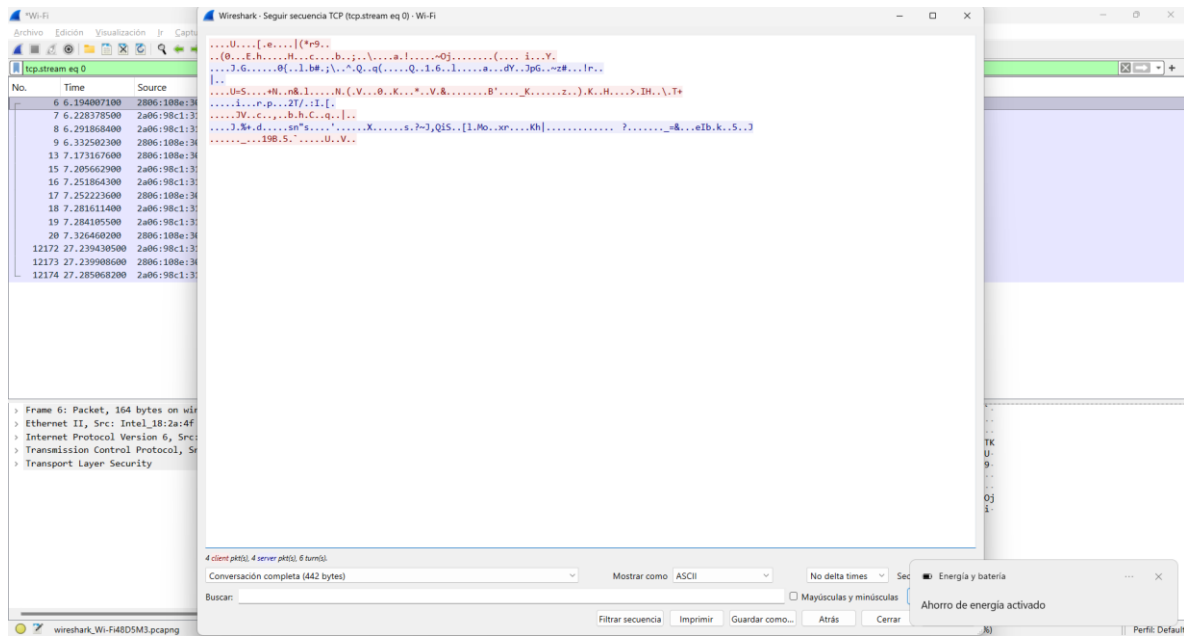
TLS-Root-2022-ECC.cer0?..U...00004..U...00*.....https://www.ssl.com/repository0...U..0.....c..U33...0..U.....0A..U...10006.4.2.0http://cr

1s.ssl.com/SSL.com-TLS-Root-2022-ECC.cer10..U.....2..X...c..U33...0..U.....0

..*H.....i..0f..1..70b(g..Y..7..f)..>..lK..x..>..s..d..b"...8

..o..F...i..M..E...JPH6.....i..*..@..Yg... ..Z.X.....M...K...

6- ¿Qué no puedo ver en https?



Al analizar el tráfico HTTPS mediante la opción *Follow TCP Stream*, se observa que la información no es legible, ya que aparece en formato cifrado. Esto impide visualizar datos sensibles como URLs, credenciales o contenido de la comunicación, lo que demuestra la seguridad del protocolo HTTPS.

HTTP:

- Se ven URLs completas
- Datos en texto plano
- Información sensible visible

HTTPS:

- Datos cifrados
- No puedes ver contenido
- Solo ves IPs y conexión